

1. (a) Advantage: could take in a lot more parameters than using the formula.

Disadvantage: slower, and depends on outside inputs like learning rate that could affect performance.

(b) MAE is more robust. Because in MSE, errors will be squared, outliers has bigger impacts, which makes MSE less robust.

(c) As $\tau \rightarrow \infty$, $w^{(i)} = e^0 = 1$.

It then becomes the normal linear regression model.

(d) (i) $O(d^p)$, because each term will interact with one another, creating d^p complexity.

(ii) It is around $O(d^{3p})$, for scanning through the data points and solving the equation.

(iii) If the samples and input parameters are relatively small, we can apply polynomial least squares. If they are large, it is better to use simple polynomial regression.

$$2. \quad \frac{1}{1+e^{-x}} \cdot e^x = \frac{e^x}{e^x+1} = p_i$$

$$1-p_i = \frac{1}{e^x+1}$$

$$\log [p_i^{y_i} (1-p_i)^{1-y_i}]$$

$$= y_i \log p_i + (1-y_i) \log (1-p_i)$$

$$= y_i \log \frac{e^x}{e^x+1} + (1-y_i) \log \frac{1}{e^x+1}$$

$$= y_i (\log e^x - \log e^x+1) + (1-y_i) - \log (1+e^x)$$

$$= y_i x - y_i \log e^x+1 + - \log (1+e^x) + y_i \log e^x$$

$$= y_i x - \log (1+e^x)$$

We put it back into a summation and add θ ,

$$\text{it becomes } l(\theta) = \sum_{i=1}^n (y_i' \theta x^i - \log (1+e^{\theta x^i}))$$

$$(b) \quad f(\theta) = \sum_{i=1}^N x^i (y^i - p(x^i))$$

$$(\theta' x^i \cdot x^i)' = p_i (1 - p_i) x^i$$

$$\Rightarrow f'(\theta) = -x^i \cdot (p_i (1 - p_i) x^i)$$

$$= -x^{i^2} p_i (1 - p_i) \quad \text{where } p_i = p(x^{(i)})$$

$$\therefore f'(\theta) = -\sum_{i=1}^N (x^i)^2 p(x^{(i)}) (1 - p(x^{(i)})).$$

$$(c) \quad \theta_{t+1} = \theta_t - \frac{f(\theta_t)}{f'(\theta_t)}$$

$$= \frac{\theta f'(\theta) - f(\theta)}{f'(\theta)}$$

extract $x_i p_i (1 - p_i)$:

$$= x_i p_i (1 - p_i) \left[-\theta x_i - \frac{y_i - p_i}{p_i (1 - p_i)} \right]$$

$$Z = \theta x_i - \frac{y_i - p_i}{p_i (1 - p_i)}$$

Homework 1

Applied Machine Learning

Date: September 24, 2026

HA455 : Harsh Agrawal

YX753 : Daniel Xing

3) Maximum Likelihood Estimation

(a) Problem Statement

You are playing a video game where you can play with four types of balls - red, blue, green and yellow. Each ball has a fixed probability to hit the goal. You play the game for a while and meticulously record the outcomes. Your objective is to determine from this record the true probabilities of the ball making its target, which we denote by goal and no_goal.

The dataset

$D = [(R, \text{goal}), (B, \text{goal}), (R, \text{no_goal}), (R, \text{no_goal}), (B, \text{goal}), (B, \text{goal}), (Y, \text{no_goal}), (G, \text{goal})]$

The dataset consists of the recorded outcomes of the ball throws. Each $(x^{(i)}, y^{(i)}) \in D$ refers to an individual throw, where $x^{(i)}$ is the ball color and $y^{(i)}$ indicates whether the ball makes its goal successfully. Each throw is represented by the ball color and its corresponding outcome.

(i) Probabilistic Model

Question: Construct a probabilistic model that captures the underlying probabilities governing the goal outcomes for each ball. Define a probabilistic model of the form $p(y | x)$ or $p(x, y)$ and its parameters, aiming to establish the probabilities associated with each outcome.

Solution :

Let,

X = color of the ball

Y = outcome of the throw of ball

Therefore,

$$X \in \{R, B, G, Y\}$$

where R , B , G and Y represent red, blue, green and yellow respectively.

$$Y \in \{\text{goal}, \text{no_goal}\}$$

Let h represent the probability of the ball making to the goal,

$$h_R = P(Y = \text{goal} | X = R)$$

$$h_B = P(Y = \text{goal} \mid X = B)$$

$$h_G = P(Y = \text{goal} \mid X = G)$$

$$h_Y = P(Y = \text{goal} \mid X = Y)$$

Therefore, the parameter vector of the probabilistic model is

$$\mathbf{h} = (h_R, h_B, h_G, h_Y)$$

Now that the probability goal =

$$P(Y = \text{goal} \mid X = x) = h_x$$

Probability of no_goal =

$$P(Y = \text{no_goal} \mid X = x) = 1 - h_x$$

Therefore, complete conditional probability model =

$$P(Y = y \mid X = x) = h_x^{\mathbb{I}(y=\text{goal})} (1 - h_x)^{\mathbb{I}(y=\text{no_goal})}$$

(ii) Learning Paradigm

Question: Classify the experiment as fitting into supervised learning, unsupervised learning or reinforcement learning paradigms.

Solution:

According to me the experiment fits into **Supervised Learning**. It contains input and corresponding output of each observation.

For each ball the output can be either goal or no_goal. They serve as the labels of the experiment and fit under **Supervised Learning**.

It is not unsupervised learning because the outcomes are explicitly provided in the dataset. It is also not reinforcement learning because there is no agent taking actions and receiving rewards or penalties from an environment.

(b) Log - Likelihood

Question: Write a formula for the log-likelihood of the dataset under the probabilistic model. Provide an intuitive argument for why we would want to optimize this objective.

Solution:

For a dataset,

$$D = \{(x^{(i)}, y^{(i)})\}_{i=1}^n,$$

The likelihood of the dataset is (assuming independent observations)

$$P(Y | \boldsymbol{\theta}) = \prod_{i=1}^n P(y^{(i)} | x^{(i)} \boldsymbol{\theta})$$

The log-likelihood is obtained by taking the logarithm of the likelihood:

$$\ell(\boldsymbol{\theta}) = \log P(Y | \boldsymbol{\theta}) = \sum_{i=1}^n \log P(y^{(i)} | x^{(i)}, \boldsymbol{\theta})$$

For the dataset

$$D = \{(R, \text{goal}), (B, \text{goal}), (R, \text{no_goal}), (R, \text{no_goal}), (B, \text{goal}), (B, \text{goal}), (Y, \text{no_goal}), (G, \text{goal})\},$$

The likelihood of the dataset is the product of the probabilities of the individual observations:

$$\begin{aligned} P(D | \boldsymbol{\theta}) &= P(\text{goal} | R)P(\text{goal} | B)P(\text{no_goal} | R)P(\text{no_goal} | R) \\ &\quad \times P(\text{goal} | B)P(\text{goal} | B)P(\text{no_goal} | Y)P(\text{goal} | G). \end{aligned}$$

The log-likelihood is the logarithm of the likelihood function:

$$\ell(\boldsymbol{\theta}) = \log P(D | \boldsymbol{\theta}) = \sum_{i=1}^n \log P(y^{(i)} | x^{(i)}, \boldsymbol{\theta})$$

We optimize the log-likelihood to find the values of the parameters that best match the outcomes we observed. A good set of parameters should assign high probability to the results that actually occurred in the dataset. For example, since all three blue ball throws resulted in a goal, the value of θ_B should be relatively high. Similarly, the red ball had one goal and two no-goals, so θ_R should reflect this pattern.

Thus, by maximizing the log-likelihood, we find the probabilities that best explain the observed results.

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$$

The logarithm is used because it converts the product of individual probabilities into a sum, making the objective easier to optimize while giving the same parameter values as maximizing the original likelihood.

(c) Maximum Likelihood Estimate

Question: Calculate the maximum likelihood estimate of the model parameters. Interpret the parameter values within the context of the ball probabilities to make the target, considering how they relate to the observed outcomes.

Solution:

The Maximum Likelihood Estimate (MLE) chooses the parameter values that maximize the likelihood of observing the given dataset. Thus,

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta})$$

Since the logarithm is a strictly increasing function, therefore maximizing the likelihood is equivalent to maximizing the log-likelihood.

Log-likelihood =

$$\ell(\boldsymbol{\theta}) = \log \theta_R + 2 \log(1 - \theta_R) + 3 \log \theta_B + \log \theta_G + \log(1 - \theta_Y).$$

Maximizing each parameter separately,

For the red ball, 1 goal 2 no-goals. Therefore,

$$\ell_R(\theta_R) = \log \theta_R + 2 \log(1 - \theta_R).$$

Taking the derivative and setting it equal to zero,

$$\frac{d\ell_R}{d\theta_R} = \frac{1}{\theta_R} - \frac{2}{1 - \theta_R} = 0.$$

$$1 - \theta_R = 2\theta_R,$$

$$\hat{\theta}_R = \frac{1}{3}.$$

For the blue ball, 3 goals:

$$\ell_B(\theta_B) = 3 \log \theta_B.$$

maximized at

$$\hat{\theta}_B = 1.$$

For the green ball, 1 goal:

$$\ell_G(\theta_G) = \log \theta_G,$$

maximized at

$$\hat{\theta}_G = 1.$$

For the yellow ball, only no-goal:

$$\ell_Y(\theta_Y) = \log(1 - \theta_Y),$$

maximized at

$$\hat{\theta}_Y = 0.$$

Therefore, the maximum likelihood estimate of the model parameters is

$$\hat{\theta} = \left(\frac{1}{3}, 1, 1, 0\right)$$

These values mean that, based on the observed data, the estimated probability of making the goal is 33.3% for the red ball, 100% for the blue ball, 100% for the green ball and 0% for the yellow ball.

(d) Limitations of MLE

Question: Recognize scenarios where this approach might yield inaccurate estimates of the true probabilities of the ball making its target. Provide at least one example and elucidate the reasoning behind potential inaccuracies.

Solution:

MLE can give inaccurate estimates when we have very few observations. For example, the green ball was thrown only once and it resulted in a goal. Based on this data, we get

$$\hat{\theta}_G = 1.$$

This means that the estimated probability of the green ball making the goal is 100%. However, this does not mean that the actual probability is 100%. If we throw the green ball many more times, some of the throws may result in a no-goal. The first successful throw could simply be due to chance.

The same problem can be seen with the yellow ball. It was also thrown only once and resulted in a no-goal, giving

$$\hat{\theta}_Y = 0.$$

The actual probability of the yellow ball making the goal could still be greater than zero. Therefore, MLE can be unreliable when the sample size is very small, since the estimate can be strongly affected by the outcomes we happen to observe.